

Delivery Company System

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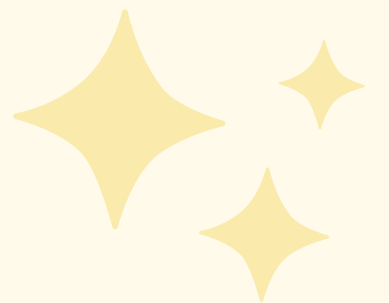
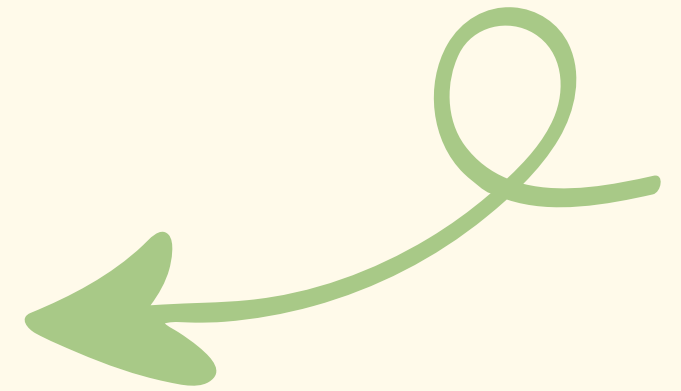
System Overview



Purpose: Manage customer orders, assign drivers, and track deliveries

Features:

- Manage customer data
- Create and process orders
- Assign drivers dynamically
- Track delivery status
- Estimate delivery time (ETA)



Main classes

01

Customer

02

Order

03

Driver

04

Company

Customer Class

Role: Represents the person placing an order (stores customer information used in orders.)

Attributes:

- name
- address
- phone_number

Concepts demonstrated:

- (private data, public getters/setters)
- Constructors/Destructors for initialization
- Exception handling for invalid contact info

Relation: Composed inside Order (an order has-a customer).



Role: Handles all data related to a delivery order.

Attributes:

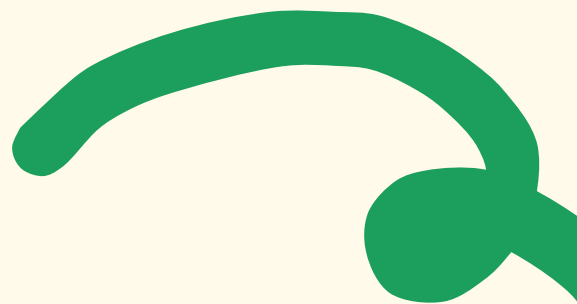
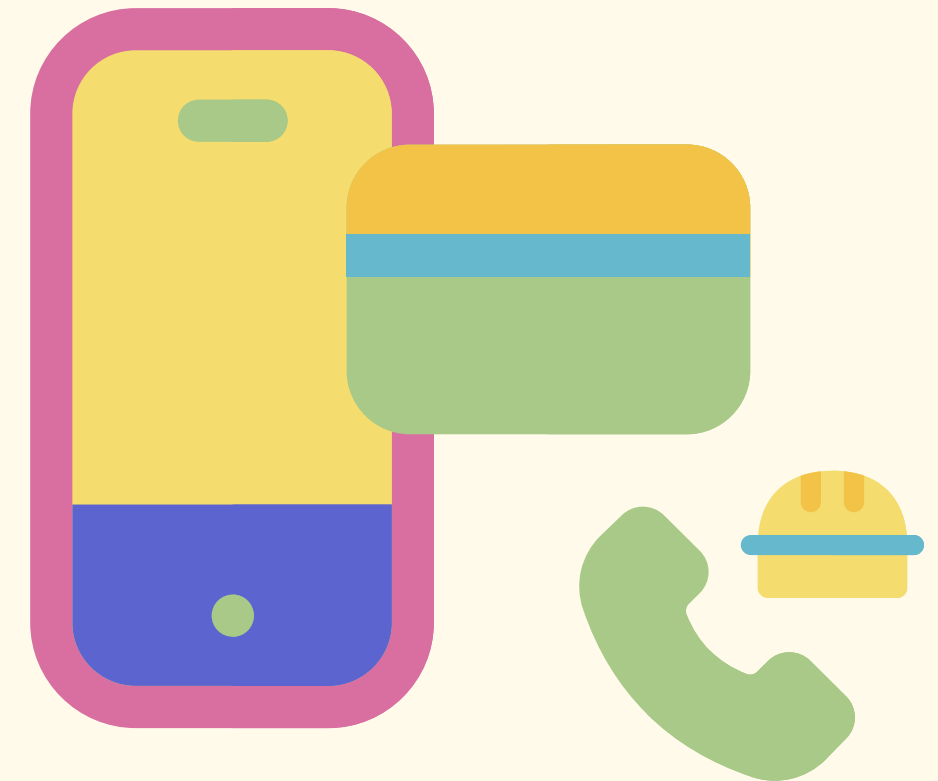
- orderID
- itemDescription
- status
- driver

Concepts Demonstrated:

- Composition (contains a customer object)
- Exception handling (invalid status or no driver)

Relation: Aggregates a Driver (pointer to driver)

Order Class



Role: Represents delivery staff.

Attributes:

- driver_ID
- name
- vehicle_type

Concepts Demonstrated:

- Inheritance (CarDriver, BikeDriver subclasses)

Relationship:

- Assigned to Orders
- Managed by Company

Driver class



Company Class

Role: Central system managing orders, drivers, and budget (control the whole system)

Attributes: Arrays of orders and drivers, budget objects.

Concepts Demonstrated:

- (manages pointers to orders and drivers)
- Exception handling (invalid IDs, unavailable drivers, budget overflow)

Relationship: Connects all classes together.

